Pegasus: Probabilistic Performance Guarantees for Agentic Provisioning of Software Services

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***Abstract*—**

***Index Terms*—Performance Guarantees, Service Quality, Gaussian Processes, Performance Modeling, Service Composition**

I. Introduction

Large Language Models (LLMs) have risen as universal interpretation and generation engines for text-based contents, introducing a major disruption for software development [7]. Traditionally, computing services were developed manually according to stakeholder requirements, and operated in close-loop controls to continuously meet operational thresholds. Within that context, LLMs appeared as a fast and cost-efficient enabler for autonomous software development and deployment, opening new branches of service research [2]: automatic generation [6], [15] and recomposition [24] of software services. The combined vision from these works is a *future internet*, where agents aim to continuously satisfy high-level user intents by generating, composing, and reconfiguring software applications and their services [1].

Traditional service-oriented architectures already provide clear roles and interfaces to operate such systems [25]: the service provider and the service consumer establish Service Level Agreements (SLAs) that specify the expected quality through measurable Service Level Objectives (SLOs). In this future internet, applications will be tailored to individual clients and their SLOs [28]. However, creating this vast space of possible application also means that potential compositions or configurations will often lack end-to-end benchmarks *prior to deployment*. It is neither possible nor cost-efficient to benchmark this search space exhaustively before offering a solution to a consumer; hence, service providers require probabilistic models for providing guarantees about the quality of different service compositions and configurations before deployment.

Applications themselves can be recomposed into systems of systems [23]; to decrease the overall complexity, individual systems (or services) must provide clear interfaces and guarantees. So suppose we need to ensure that one system's latency is below a threshold, it is not enough to *estimate* that the tail latency (e.g., the P_{99} percentile) will be below the threshold, but the application must *guarantee* this with a certain probability. To avoid SLA violations, providers could simply give increasingly conservative performance estimates, which could only be fulfilled by provisioning more resources, increasing the price for the consumer, and wasting the provider's resources. The challenge is thus to make the performance guarantee as *tight* as possible. For a service composition S, also called an application, with topology $\{s_1 \rightarrow s_2 \rightarrow \dots\}$, a provider needs to be able to accurately answer the following questions:

- What performance could we guaranteed for this application, before conducting any end-to-end benchmark?
- What performance could we guarantee if we replaced a service (e.g., s_1) with another one (e.g., s_x) that was not benchmarked end-to-end in this exact application?
- What performance could we guarantee if we reconfigured an individual service (e.g., s_1) to a configuration that was not benchmarked end-to-end in this application?

While composition and configuration are operationally decoupled—services can be reconfigured without altering the topology, and vice versa—they are logically entangled: estimating composite performance requires understanding individual service dynamics under limited empirical observations. Optimization techniques like branch-and-bound can reduce the candidate space, but evaluating candidate solutions remains the core bottleneck. Learning-based optimizers must contend with varying degrees of empirical evidence, ranging from well-observed service instances to completely unseen configurations and compositions.

TODO: Add summary paragraph introducing the contribution/solution: State the gap in existing work (e.g., deterministic code contracts vs. black-box microservices) and summarize the proposed PPG framework combining SVGPs and Stochastic Network Calculus.

II. Preliminaries

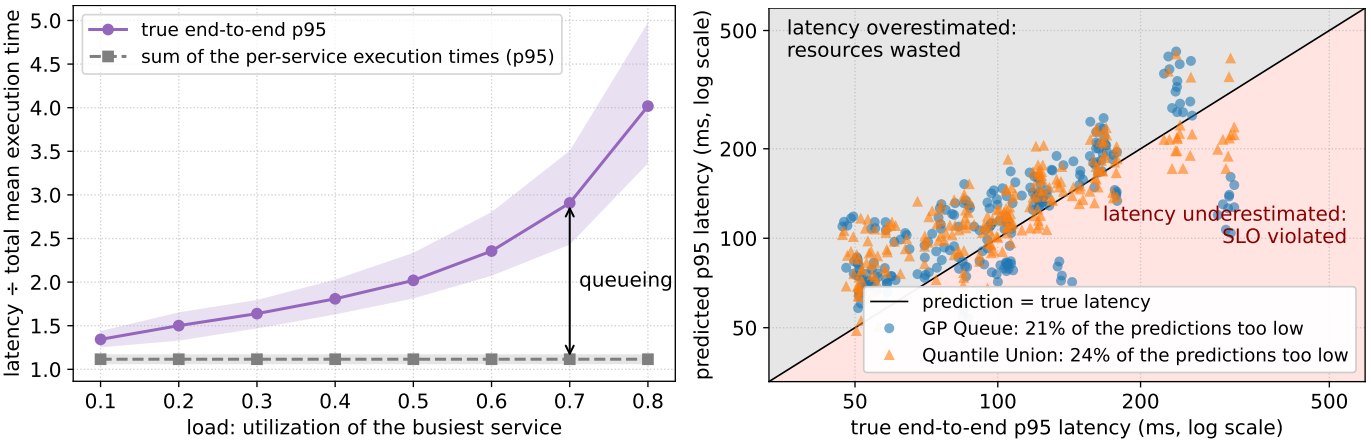
A. Motivating Example

Consider a customer that requests a real-time drone monitoring application, which can be structured as a four-stage pipeline: (1) video stream ingestion from the drone, (2) object detection along the trajectory, (3) trajectory adjustment calculation, and (4) telemetry reporting to an external traffic planner. Logically, this application is composed of multiple connected microservices $S = \{s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow s_4\}$.

Each service in this application can be configured in terms of allocated hardware resources (e.g., CPU cores), the number

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(a) Per-service execution times do not add up to the end-to-end tail latency. (b) An accurate prediction is not a guarantee and can err in both directions.

Fig. 1. Composing per-service performance models does not yield an end-to-end guarantee (three computer vision services; measured execution times replayed through FIFO queues under Poisson arrivals). **(a)** True end-to-end P_{95} and the sum of the per-service P_{95} execution times, both divided by the total mean execution time of the chain, against the utilization of the busiest service; lines show the median and bands the interquartile range over the configurations, and the gap between them is the queueing time. **(b)** Predicted versus true P_{95} at utilization 0.5 (5 training sets, 240 predictions per method) for a queueing formula on learned execution times (*GP Queue* [4]) and learned per-service latency quantiles (*Quantile Union*), both trained on individual services only; the legend gives the share of predictions below the true latency.

of service replicas, and internal service parameters (e.g., object detection model resolution or batch size). While all these configuration knobs influence execution time, the customer is solely interested in end-to-end *performance* (e.g., total latency ≤ 100 ms) and overall *cost* (e.g., minimize resources).

Furthermore, external systems like the traffic planner pose strict workload bounds, e.g., updates must arrive consistently every 100 ms. Adhering to these bounds in distributed systems typically relies on reporting high-percentile tail latencies (e.g., P_{90} or P_{99}). However, the specified application topology and configuration have not been executed and benchmarked end-to-end so far; the provider only has measurements of the individual services, collected in isolation or in another existing applications. The following section shows why these measurements alone do not yield the required guarantee.

B. Research Questions

Figure 1 shows why naive analysis and composition of service observations fail to guarantee performance, and motivates the two research questions that this paper addresses. The application is a chain of three computer vision services from our testbed in Section V, each configurable in its number of service replicas and quality knobs (e.g., input resolution, model size). For each of the 50 measured configurations of the chain, we replay the per-item execution times measured at every service through FIFO queues, with Poisson arrivals at a rate that keeps the busiest service busy for a given share of the time (its utilization). All models are trained only on runs of the services alone or in other chains, never on this chain.

Composing Tail Latencies under Load. The end-to-end latency of an item is the sum of its execution times *plus* the time it waits in the queue of every service. This waiting time cannot be observed at an individual service: it depends on how irregularly items enter the application (the external arrival process), on how each service releases items to the next one (the internal composition), and on how close each service runs to its capacity (the load). Figure 1a shows that the sum of the per-service P_{95} execution times underestimates the end-to-end P_{95} at every load, and that the gap grows significantly with increasing load. Benchmarking a candidate composition can capture end-to-end latency, including queuing time [21], but, as argued in the introduction, the space of possible compositions and configurations is too large to benchmark exhaustively. **Question #1:** The open research question is thus how to derive the end-to-end tail latency of a composition, including its queueing time, purely from observations of individual services and a workload description.

Guarantees for Unseen Configurations. An SLO requires a latency bound that is exceeded with at most a given probability; however, performance models are fitted to be accurate [8], [31], i.e., *close* to the measured value on average, but not necessarily above it. For a configuration that was never observed, such a prediction can err in either direction: an overestimated latency wastes resources, an underestimated one violates SLOs. Figure 1b shows that two predictors that account for queueing, trained only on observations of individual services, overestimate the latency for most unseen configurations but underestimate it for a sizable share. Even quantile predictors [14] give no indication of whether a configuration is well covered by observations or far away from all of them. **Question #2:** The open research question is therefore how to turn limited evidence about each service into an end-to-end bound that holds with a stated probability, and becomes more conservative where evidence is missing.

C. Background and Related Work

TODO: To improve the flow, we must split this section: background stays here, and RW goes to the paper's end

We have grouped related work by how it obtains the performance of an application: (1) analytical models derive it from

a formal description of the system, (2) learned models fit it to measurements of the entire application, and (3) compositional models assemble it from models of the individual services. For each group, we explain *how* it models performance and *why* it leaves the presented research challenges open.

1) Analytical Performance Models: Queueing theory is a classical tool for relating workload and capacity to response time [16]. For instance, Gandhi et al. [13] model the performance of individual servers as M/G/1 queues[1], and use this model for finding a power allocation that minimizes overall response time. Layered Queueing Networks [10] extend queueing networks to software services that hold one resource while waiting for another, as in nested client–server calls. Both target mean response times in steady state rather than tail probabilities (Challenge #2). Network calculus bounds delays instead of averaging them: Duan [5] describes network and compute services through latency–rate capability profiles, merges these profiles into one end-to-end profile, and derives the maximum delay of the composite service. This bound is deterministic, i.e., it covers the worst case and thus becomes overly conservative for services with variable execution times; Stochastic Network Calculus (SNC) [9], on which we build, instead bounds the *probability* of exceeding a given delay. SNC assumes that the MGFs of arrivals and service exist; an extension to heavy-tailed traffic replaces them by power-law envelopes at the price of considerably looser bounds [22].

Closest in spirit, Iyer et al. [17] compute *performance contracts* for software network functions without executing them: symbolic analysis of the source code yields conservative upper bounds on the memory accesses and execution cycles per packet, expressed as functions of workload-dependent variables; contracts for chains of network functions are computed recursively. As with worst-case execution time analysis [33], this requires access to the source code, whereas microservices are typically deployed as opaque containers. Moreover, these contracts bound the processing effort per packet, not waiting time that accumulates under load (Challenge #1).

Common to all analytical models is that their parameters (e.g., service rates, branching probabilities, or code paths) must be supplied for the exact system under study. None of them expresses how reliable a prediction is for a configuration that was never measured (Challenge #2).

2) Learning Models of Entire Applications: A second line of work treats the application as one big black box and learns its end-to-end performance from empirical measurements; we call this the *top-down* approach. *Ernest* [31] fits a small parametric model to short profiling runs of an analytics job on samples of the input data, and extrapolates the running time to larger datasets and clusters. *CherryPick* [3] replaces the fixed model structure with a Gaussian Process (GP) and uses its confidence intervals to choose the next cloud configuration to benchmark; the search stops as soon as the model can identify the best configuration. *Microscaler* [36] similarly combines Bayesian optimization (BO) with a heuristic to find the number of instances at which a microservice meets its SLA. Production systems are more conservative: Google's *Autopilot* [27] sets resource limits from moving-window statistics of past usage plus a safety margin, optionally choosing among many such window models by their historical cost.

Top-down models are simple in the sense that they require no knowledge of the application's internal structure: a single model maps the configuration to the end-to-end performance. This is also their limitation: replacing a service or changing the topology changes the modeled function itself, so the end-to-end performance must be benchmarked again (Challenge #1). Furthermore, where uncertainty is modeled at all, as in *CherryPick*, it steers the search towards a good configuration but is not converted into a latency guarantee (Challenge #2).

3) Compositional Models: The *bottom-up* alternative predicts the quality of a composition from its constituent services, which has a long tradition in service-oriented computing. Gallotti et al. [12] translate activity diagrams into stochastic models for a probabilistic model checker; the execution time and reliability of each service are annotations supplied at design time. Wu et al. [35] use a hidden Markov model to predict whether a composition of Cloud services will satisfy the requested QoS, and Fanjiang et al. [8] evolve time-series predictors that forecast the QoS of web services. Geebelen et al. [14] come closest to a guarantee: a Petri-net simulation merges the monitored response times of the elementary services into a time series of the composition, and a kernel-based quantile estimator with an online offset correction predicts the quantile named in the SLO, such that the violation frequency converges to the agreed level. None of these works, however, learns QoS as a function of resource allocation, service configuration, and arrival rate. They can assess compositions of services as currently deployed, but not a composition that involves a configuration without observations; likewise, the guarantee of Geebelen et al. holds asymptotically in the number of observations (Challenge #2).

Fluxion [21] is the closest learned counterpart to our work. It trains one model per service (e.g., a deep neural network or a GP) that maps the service's configuration knobs, its observed states, and the latency percentiles of its downstream services to a latency percentile of the service itself. These models are chained into an inference graph that mirrors the service dependencies; after a service is scaled or replaced, only the model of that service is retrained. However, the inference graph propagates point estimates: Fluxion is evaluated by its mean absolute error, and its authors note that prediction errors accumulate along the graph in a way that is not trivial to derive analytically. A predicted P_{90} can therefore err in either direction, and the output does not reveal any uncertainty about whether a service model was queried at a configuration it has never seen (Challenge #2).

D. Summary

In summary, analytical models compose rigorously but need access to parameters (e.g., source code); top-down approaches can wrap around black-box applications, but this prevents

[1]Note that M implies a *Markovian* (or Poisson) arrival process, G a *General* service time distribution, i.e., described through a mean and variance, and $c = 1$ indicates a single server. *Pegasus* supports a wider range of arrival processes and multi-server queues with $c \geq 1$, but assumes a sub-Gaussian service process (e.g., Gaussian or bounded execution times).

generalization to unseen topologies; and existing compositional models propagate predictions without any degree of uncertainty. *Pegasus* (our approach) combines the strengths of these approaches: it learns the execution time distribution of each black-box service from local observations, including the epistemic uncertainty for unseen configurations, and composes per-service models into a probabilistic bound on the end-to-end latency for unseen service compositions.

III. Problem Definition

A. System Model

We model a service composition S, also called an application, as a Directed Acyclic Graph (DAG) with available microservices (vertices) $V = \{s_1, s_2, \ldots, s_n\}$, where directed edges $E \subseteq V \times V$ represent the paths along which items (e.g., requests or data frames) are passed on. For a sequential pipeline, this simplifies to $S = \{s_1 \to s_2 \to \cdots \to s_n\}$ with $n = |S|$ services.

Services. Each service $s_i \in V$ consists of c_i instances, where all instances share one configuration $x_i = \langle r_i, \kappa_i \rangle$, where

- $r_i \in \mathcal{R}_i$ are the global resource allocations (e.g., CPU quota, memory limit) across all service instances,
- $\kappa_i \in \mathcal{K}_i$ are its software configuration knobs (e.g., model size, data quality) unified across all instances, and
- $c_i \in \mathbb{N}$ is the number of servers/workers for s_i.

A service processes up to c_i items at a time, one per worker, in the order of arrival; items that arrive while the service is busy wait in its queue.

Latency. The *execution time* Y_i of s_i is the time (in seconds) from the start to the end of processing one item. Due to runtime variability (e.g., thread scheduling, input size), Y_i is a random variable whose unknown probability distribution $P_{Y_i|x_i}$ depends on the service configuration x_i. The *latency* of an item at s_i is its *waiting time* in the queue plus its execution time: $W_i = Q_i + Y_i$. Hence, for K services, the end-to-end latency D_S is the sum of their latencies,

$$D_S = \sum_{i=1}^{K} W_i \ . \tag{1}$$

Data Collection. For each service, empirical evidence is available as a sparse dataset of execution times y_i observed at N_i configurations, $\mathcal{D}_i = \{(x_i^{(j)}, y_i^{(j)})\}_{j=1}^{N_i}$. These samples are collected while the service ran in isolation (benchmarked on their own) or within another composition. Thus, only a small share of all possible service compositions $(V \times V)^n$ are benchmarked end-to-end; likewise, large regions of their configuration space $\mathcal{R}_i \times \mathcal{K}_i$ remain unobserved.

Workload. Items enter the composition at its first service according to an *external arrival process*, which we describe by the gap T (in seconds) between two consecutive arrivals. We define the mean arrival rate λ (items per second) as

$$\lambda = \frac{1}{\mathbb{E}[T]}, \tag{2}$$

and the variability of the arrivals C_a^2 through the squared coefficient of variation as

$$C_a^2 = \frac{\mathrm{Var}(T)}{\mathbb{E}[T]^2}. \tag{3}$$

An item that leaves a service immediately enters the next one, so every downstream service receives the departures of its predecessor.

TODO: Place the utilization formula somewhere

$$\rho_i = \frac{\lambda \times m_i}{c_i} \tag{4}$$

Probabilistic Performance Guarantees. Let D_S denote the end-to-end latency in a composition S under the configuration vector $\mathbf{x} = (x_1, \ldots, x_K)$, i.e., the time an item spends from its external arrival until it leaves the last service. Given a maximum permissible violation probability $\epsilon \in (0, 1)$ specified in an SLO, a latency bound d_S must satisfy

$$\mathbb{P}\left(D_S > d_S\right) \leq \epsilon, \tag{5}$$

which means that d_S is an upper bound on the $(1-\epsilon)$ quantile of the end-to-end latency.

TODO: Might need an example with some values

Because the execution-time distributions are modeled using sparse observations $\mathcal{D}$, the latency bound in Eq. (5) contains noise and uncertainty around poorly explored configurations. A *Probabilistic Performance Guarantee* (PPG) therefore has two parameters: the violation probability ϵ, which covers the variability between items, and a confidence level $\gamma \in (0, 1)$, which covers the uncertainty of the learned models. A PPG is a latency bound $d_S(\mathbf{x}, \epsilon, \gamma)$ for which Eq. (5) holds with confidence γ.

B. Problem Statement.

Given a composition S, a configuration vector $\mathbf{x}$, an external arrival process λ, C_a^2, a violation probability ϵ, and a confidence level γ, derive the PPG $d_S(\mathbf{x}, \epsilon, \gamma)$ in a *zero-shot manner*, which means purely from the per-service datasets $\{\mathcal{D}_1, \ldots, \mathcal{D}_K\}$ and without any end-to-end measurement of S. The bound should be as tight as possible, since every unit of slack translates into over-provisioned resources, and account for the queuing at every service.

C. Foundational Assumptions

To obtain performance bounds for black-box service compositions without end-to-end measurements, our framework makes the following assumptions about the domain:

1) *Performance Isolation:* Services run in isolated execution environments (e.g., containers with strict CPU and memory limits), and the CPU quota per service suffices for executing all its workers without throttling. The execution time of a service is therefore neither affected by co-located services nor by its own load.
2) *Multi-Server FIFO Services:* Every service has c_i workers that share one unbounded queue and process items in the order of their arrival (i.e., FIFO). For constructing

the bound, we assume that a worker receives every c_i-th item (*cyclic assignment*), turning it into a single-server queue whose gap between two items is the sum of c_i consecutive inter-arrival times. Cyclic assignment can only increase the waiting time [34]; hence, upper bounds under cyclic assignment also hold for shared queues.

3) *Load-Independent Execution Times:* The distribution of the execution time Y_i depends only on the configuration x_i; thus, Y_i depends neither on the arrival process nor on the services before s_i, and the execution times of different items at s_i are independent. This allows execution times observed in isolation, or in a different composition, to be reused for unseen compositions.
4) *Sub-Gaussian Execution Times:* The moment-generating function (MGF) of the execution time Y_i is bounded by that of a Gaussian with mean $m_i = \mathbb{E}[Y_i]$ and variance proxy σ_i^2, both depending on x_i, for all $\theta > 0$ (in 1/s):

$$M_{Y_i}(\theta) = \mathbb{E}\left[e^{\theta Y_i}\right] \leq \exp\left(\theta\, m_i + \frac{\theta^2}{2}\,\sigma_i^2\right). \qquad (6)$$

This holds, e.g., for Gaussian and for bounded execution times; heavier tails are discussed in Section VI-A.
5) *Renewal Arrivals:* The gaps T between external arrivals are independent, identically distributed (i.i.d.), and independent of the execution times, and their Laplace transform $\mathbb{E}[e^{-\theta T}]$ is known. This covers, for instance, Poisson arrivals, periodic arrivals with jitter, and gamma-distributed gaps of any C_a^2.
6) *Arrivals Inside the Composition:* A service that is not the first of the composition is bounded as if it received the external arrival process, refined by the minimum spacing that its predecessor enforces between two departures (Section IV-B). This is conservative when the external arrivals are at least as variable as the execution times, as with Poisson arrivals; for nearly periodic arrivals combined with highly variable execution times and high utilization, the departures of a service can be burstier than the external arrivals.

Independence between the execution times of an item at *different* services is not assumed in general; it is only required by one of the two presented framework variants (Section IV-C).

IV. Pegasus Framework

Given limited observations from individual services, our framework, called *Pegasus*, allows bounding the performance of any potential composition and configuration of these services. This bound—the PPG—presents a probabilistic performance guarantee for the end-to-end latency according to an external arrival process and a desired confidence level. Our methodology consists of three steps, as shown in Figure 2: (1) learning a probabilistic model of the execution time of each service, (2) deriving a latency bound for each service from its execution time model and expected arrival process, and (3) composing the service bounds into a performance guarantee for a complete application. Only the first step requires training data for modeling the performance of individual services; the arrival processes and the composition enter analytically in the second and third step. This separation makes the guarantee available for compositions that were never benchmarked.

In the following subsections, we explain the three steps of the *Pegasus* framework in more detail. For each step, we also provide technical details on how Pegasus was implemented. The implementation is publicly accessible on the Pegasus Repository; Appendix 3 contains an extended description of the repository and how to use Pegasus for a custom dataset.

A. Learning per Service Performance Models

The first step trains, for each service s_i, a probabilistic model of the execution time Y_i for processing an item, given a service configuration x_i. Two constraints apply to this step: (1) the performance of an individual service instance does rarely scale linearly with increasing resources—with sufficient resources, the performance saturates [3]—and (2) services live in containers, whose source code cannot be analyzed to make assumptions about their functional behavior (unlike related works in Section II-C1). Gaussian Processes (GPs) satisfy these two constraints: first, they are non-parametric and do not impose any structure on the learned function; second, unlike point-estimate models (e.g., neural networks), they quantify the uncertainty of predictions, which protects the bound at unobserved configurations outside the training set.

We train one GP per service on its dataset $\mathcal{D}_i$. For any configuration x_i, which may not be contained in the training data $\mathcal{D}_i$, the GP provides three quantities:

- $m_i(x_i)$—the predicted mean execution time;
- $\sigma_{\mathrm{ep},i}(x_i)$—the *epistemic* uncertainty around the predicted mean execution time, which is high under scarce coverage and shrinks with additional observations; and
- $\sigma_{\mathrm{al},i}(x_i)$—the *aleatoric* uncertainty as a measure of the noise or scatter of predictions around the mean; it is induced by randomness in the execution-time distribution, and hence, does not shrink with more observations.

In the following two steps of *Pegasus*, we distinguish these two types of uncertainty to differentiate between areas in the configuration space that are simply not covered, and areas that are inherently noisy. In the remainder of this section, we omit the argument x_i where it is clear from the context.

Implementation. Each service's performance model is implemented through an exact GP in scikit-learn [26] with a scaled Matérn-5/2 kernel, one length scale per input, and an additive white-noise term, fitted on standardized inputs by maximizing the marginal likelihood (five restarts). This means that $\sigma^2_{\mathrm{ep},i}$ is the posterior variance of the latent function and $\sigma^2_{\mathrm{al},i}$ the fitted noise level. Configuration knobs act multiplicatively on the execution time (e.g., a larger model at a higher resolution), hence, we fit the GPs on the logarithm of their target.

B. Modeling Execution Time and Load per Service

The second step turns the performance model of s_i and the arrival process into the two quantities that both composition variants build on: the MGF M_i of the execution time and the load ϕ_i of the service. Both variants operate on MGFs: Chernoff's bound $\mathbb{P}(Y > d) \leq e^{-\theta d} M_Y(\theta)$, $\theta > 0$, controls

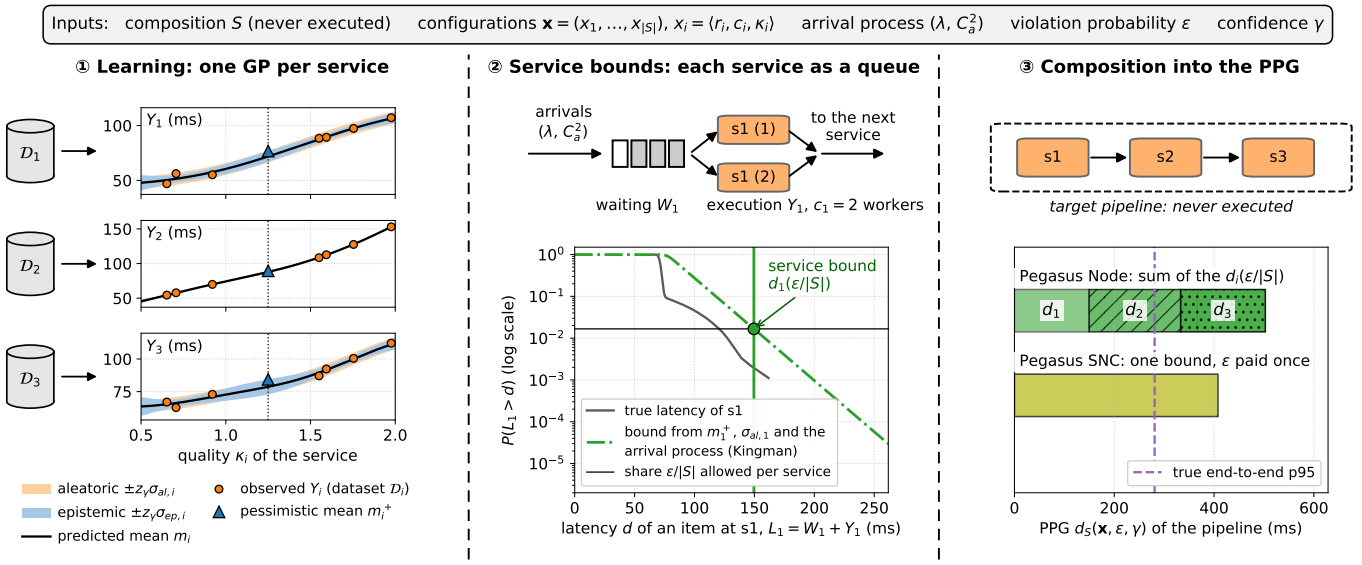


Fig. 2. Pegasus for a application that has never been executed. ❶ One GP per service, trained only on that service's execution times $\mathcal{D}_i$, yields a pessimistic mean m_i^+ that grows with the epistemic uncertainty; ❷ Each service is bounded as a FIFO queue with c_i workers, which gives the service bound $d_i(\epsilon/K)$; ❸ The service bounds are composed into the PPG $d_S(\mathbf{x}, \epsilon, \gamma)$, either as their sum (Pegasus Node) or as one application-wide bound (Pegasus SNC).

the tail of a time Y, the MGF of a sum of independent times is the product of their MGFs, and the Laplace transform of the gaps (Assumption 5) is their MGF at $-\theta$.

Pessimistic Execution-Time Model. The aleatoric scatter is independent between items, so it partly averages out over a busy period. The epistemic error does not: suppose the model underestimates the mean by 5 ms, *every* item takes 5 ms longer, and a busy period of 100 items grows by 500 ms. Adding both variances per item would therefore underestimate the queue. Instead, we let the epistemic uncertainty shift the mean and keep only the aleatoric variance per item:

$$m_i^+ = m_i + z_\gamma\, \sigma_{\mathrm{ep},i}, \qquad Y_i \sim \mathcal{N}\big(m_i^+,\, \sigma_{\mathrm{al},i}^2\big)\,, \tag{7}$$

where z_γ is the γ-quantile of the standard normal distribution, e.g., for $\gamma = 0.95$, the $z_\gamma \approx 1.64$. According to the model, the true mean execution time lies below the *pessimistic mean* m_i^+ with probability γ. This is where the confidence level of the PPG enters: following Eq. (6), the MGF of the execution time is $M_i(\theta) = \exp(\theta m_i^+ + \frac{\theta^2}{2}\sigma_{\mathrm{al},i}^2)$. Chernoff's bound $\mathbb{P}(Y > d) \le e^{-\theta d} M_Y(\theta)$ holds for every $\theta > 0$; θ (in 1/s) is a free parameter that can be optimized to make the bound as tight as possible.

Load. The arrivals enter through the Laplace transform of the gaps, $\mathbb{E}[e^{-\theta T}]$; for Poisson arrivals at rate λ, for instance, $\mathbb{E}[e^{-\theta T}] = \lambda/(\lambda+\theta)$. Under cyclic assignment (Assumption 2), each worker of s_i is a single-server queue whose gap between two items is the sum of c_i gaps T, which gives the load

$$\phi_i(\theta) = M_i(\theta)\, \mathbb{E}\big[e^{-\theta T}\big]^{c_i}. \tag{8}$$

Both variants need $\phi_i(\theta) < 1$ for some $\theta > 0$; otherwise, the service is overloaded and has no finite bound. Since ϕ_i uses m_i^+, this can also happen for a stable service whose mean is too uncertain (i.e., less explored).

Implementation. Pegasus implements Poisson arrivals, gamma-distributed gaps with any C_a^2, where $\mathbb{E}[e^{-\theta T}] = (1+\theta C_a^2/\lambda)^{-1/C_a^2}$, and periodic arrivals with uniform jitter; further renewal processes only need their Laplace transform. Both variants evaluate θ on a logarithmic grid of 4000 values between 0.1 and $10^5\,\mathrm{s}^{-1}$, which can only loosen their bounds. For the log-scale GPs of the testbed, the shift of Eq. (7) is relative, $m_i^+ = m_i\, e^{z_\gamma \sigma_{\mathrm{ep},i}}$, and the predicted spread enters without a margin of its own.

C. Composing Bounds for Unseen Applications

The third step composes the services models into a performance bound of the entire application S. We provide two variants for achieving this, which trade off assumptions against tightness. *Pegasus SNC* pays $-\ln\epsilon$ once for the whole application ("pay bursts only once") instead of bounding every service at ϵ/K, and needs only the external arrivals (no Assumption 6); it is therefore the tighter variant for long applications. *Pegasus Node* is required, or preferable, if (1) a service has several workers, since *Pegasus SNC* needs items to leave every service in arrival order; (2) the execution times of an item are correlated between services, which the union bound tolerates, whereas *Pegasus SNC* needs Hölder's inequality, scaling the variance of s_i by $p_i \ge 1$, $\sum_i 1/p_i = 1$ (ablation); or (3) one service is close to saturation, which loosens only its own bound instead of the shared θ of the whole application.

Composition per Service (*Pegasus Node*). Pegasus Node bounds every service on its own, from its own model, the arrival process, and at most its predecessor, and adds the bounds. The waiting time of an item at a FIFO service is the largest backlog of work that the items before left behind. Since the gaps are independent (Assumption 5), Kingman's martingale bound [19] states that this backlog has an expo-

nentially decaying tail: for any waiting-time threshold $w \geq 0$ (in s),

$$\mathbb{P}(Q_i > w) \leq e^{-\theta_i^* w}, \quad \theta_i^* = \sup\{\theta > 0 : \phi_i(\theta) \leq 1\}. \quad (9)$$

The *decay rate* θ_i^* (in $1/\text{s}$) shrinks with everything that lets the queue grow: a larger mean or variance of the execution time, a higher arrival rate, or more variable arrivals. The waiting time of an item depends only on the items before it, and is therefore independent of its own execution time. Combining Eq. (9) with the Gaussian execution time of Eq. (7) gives the tail of the latency in closed form,[2]

$$\mathbb{P}(W_i > d) \leq \bar{\Phi}\left(\frac{d - m_i^+}{\sigma_{\text{al},i}}\right) + e^{-\theta_i^*(d-m_i^+)+\frac{1}{2}(\theta_i^*\sigma_{\text{al},i})^2}\, \Phi\left(\frac{d - m_i^+ - \theta_i^*\sigma_{\text{al},i}^2}{\sigma_{\text{al},i}}\right), \quad (10)$$

where Φ is the standard normal distribution function and $\bar{\Phi} = 1 - \Phi$. The first term is the probability that the execution time alone exceeds d; the second term adds the waiting time. The *service bound* $d_i(\epsilon)$ is the smallest d for which the right-hand side is at most ϵ.

Services Inside an Application.

TODO: @Milos: I implemented this minimum spacing to squeeze a bit more tightness from the Pegasus bounds; however, this only gives an improvement of around 5% while making the whole method more complicated. Additionally, it can't be used for multiple workers, so we might simply skip it and move to future work.

Only frontend services receive the external arrivals; every subsequent service s_i receives the departures of its predecessor s_{i-1}. We generally bound s_i with the external arrivals (Assumption 6), and tighten the bound if s_{i-1} has a single worker: its departures are then at least one execution time of s_{i-1} apart, so a worker of s_i, which receives every c_i-th item, sees gaps of at least c_i such execution times. Applying Kingman's bound to the execution times of s_i against this *minimum spacing* gives a second decay rate

$$\theta_i^{\text{s}} = \frac{2\left(c_i\, m_{i-1}^- - m_i^+\right)}{\sigma_{\text{al},i}^2 + c_i\, \sigma_{\text{al},i-1}^2}, \text{ with} \quad (11)$$

$$m_{i-1}^- = m_{i-1} - z_\gamma\, \sigma_{\text{ep},i-1}, \quad (12)$$

which is positive only if $c_i\, m_{i-1}^- > m_i^+$, i.e., if s_{i-1} is slower even with its *optimistic* mean m_{i-1}^- (a faster predecessor spaces the items less). After a predecessor with several workers, $\theta_i^{\text{s}} = 0$. Eq. (10) uses the faster rate, $\max(\theta_i^*, \theta_i^{\text{s}})$; Eq. (11) treats the execution times of consecutive items at the two services as independent.

[2]Eq. (10) is the only step that uses the Gaussian shape of Y_i; all other steps only need the sub-Gaussian bound of Eq. (6). For a sub-Gaussian but non-Gaussian Y_i, splitting $d = d_Y + d_Q$ and bounding $\mathbb{P}(Y_i > d_Y) \leq e^{-(d_Y - m_i^+)^2/(2\sigma_i^2)}$ and $\mathbb{P}(Q_i > d_Q) \leq e^{-\theta_i^* d_Q}$ separately gives a slightly looser bound.

Union Bound. An item can only exceed the sum of the service bounds if it exceeds at least one of them. Hence, if every service is bounded at the violation probability ϵ/K,

$$d_S = \sum_{i=1}^{K} d_i\left(\frac{\epsilon}{K}\right) \quad \text{implies } (\Rightarrow) \quad (13)$$

$$\mathbb{P}\left(D_S > d_S\right) \leq \sum_{i=1}^{K} \mathbb{P}\left(W_i > d_i\left(\frac{\epsilon}{K}\right)\right) \leq \epsilon. \quad (14)$$

This requires no knowledge of how the latencies at different services depend on each other, so it remains valid if an item that is slow at one service is also slow at the next (e.g., a large video frame), and although consecutive queues are coupled.

Composition per Application (*Pegasus SNC*). Stochastic Network Calculus (SNC) [9] composes the *service* of multiple nodes instead of their bounds. Let $A(n)$ be the arrival time of item n and $D_i(n)$ its departure from s_i, with $D_0(n) = A(n)$. A single-worker FIFO service obeys

$$D_i(n) = \max\{D_{i-1}(n),\, D_i(n-1)\} + Y_i(n). \quad (15)$$

Unrolling Eq. (15) writes $D_S = D_K(n) - A(n)$ as a maximum over (item, service) paths; a union bound over them, one Chernoff step, and a geometric series per service yield, for every $\theta > 0$ with $\phi_i(\theta) < 1$ (Eq. (8) with $c_i = 1$),

$$\mathbb{P}\left(D_S > d\right) \leq e^{-\theta d} \prod_{i=1}^{K} \frac{M_i(\theta)}{1 - \phi_i(\theta)}. \quad (16)$$

Each factor is the execution-time MGF of s_i, inflated by the queue that the load $\phi_i(\theta)$ causes. Setting the bound to ϵ and optimizing θ jointly over the models of all services gives

$$d_S = \min_{\theta > 0} \frac{1}{\theta}\left[\sum_{i=1}^{K} \ln \frac{M_i(\theta)}{1 - \phi_i(\theta)} - \ln \epsilon\right]. \quad (17)$$

Implementation. θ_i^* is the largest grid value that satisfies Eq. (9), and $d_i(\epsilon)$ is found by bisection on Eq. (10), returning the upper end of the final interval; for log-scale GPs, $m_i^- = m_i\, e^{-z_\gamma \sigma_{\text{ep},i}}$. Both variants are closed-form expressions that need neither simulation nor sampling, and their cost grows linearly with K. For the Hölder ablation, the exponents are tuned per configuration and θ by a fixed-point iteration on $1/p_i \propto \sigma_{\text{al},i}/\sqrt{1 - \phi_i(\theta)}$. Both variants extend to fork-join DAGs, whose end-to-end latency is the maximum over the P leaf-to-root paths: each path is bounded as an application at ϵ/P (*Pegasus SNC*), or each of its services at $\epsilon/\sum_p K_p$, with K_p services on path p (*Pegasus Node*), and the DAG bound is the largest path bound.

V. Evaluation

TODO: Pegasus SNC does not work for multi-server setups, so we need to remove it from all the multi-server experiments. That's ok, because the only one where it clearly wins is B3, and there we can evaluate a single-server scenario.

TABLE I
SERVICES AND KNOBS PER APPLICATION.

App.	Services	Knobs
Vision	object/face/edge/motion/QR detector, transformer, compressor, denoiser	160–640 px, model
Audio	resampler, denoiser, features, voice detector, pitch tracker, transcriber, encoder	2–12 unit??
Lidar	downsampler, ground remover, clusterer, normals, registrar, visualizer, encoder	6–40 unit??

A. Experimental Setup

Service Generation. To evaluate generalization across structural dimensions, we evaluate unseen service configurations, unseen service compositions, and their combination. This requires an ample set of services and service compositions; for this, we combine an existing set of microservices and configuration interfaces from prior work [29] and augment it through LLM-based code generation.

TODO: Name the LLM and point to appendix 1.

Applications. We build three applications (Table I), each with five chains of 3–6 services; every service appears in 2–3 chains. All services of a chain share one data quality per run. Each service has one worker and a FIFO queue. ($c_i > 1$ *only in synthetic data.*)

TODO: measure $c_i > 1$

Testbed. Services run in Docker with 1–2 cores on one host (AMD Ryzen 9 7950X, 128 GB RAM). Each run draws one configuration uniformly at random. An 8 s probe sets the Poisson arrival rate to 50% utilization of the slowest service; after a 10 s warm-up, the run records the waiting and execution time of every item at every service. We run 150 runs per chain and keep the 75–106 without CPU throttling, which would make execution times load-dependent (Assumption 3). (*Most runs have only a few hundred items.*)

TODO: longer runs; GPU?; regenerate Fig. 1 on this data

Training Data. By default, every service is trained on 25 runs of its isolated benchmark (10 configurations, each at utilizations 0.1–1.0), never on the evaluated chain. Each experiment uses three seeds, which draw the training runs. B3, B4, and C2 need settings the testbed has not measured; they use generated services and a simulator of FIFO queues.

Baselines. We compare *Pegasus* against five baseline methods. Like *Pegasus*, each baseline is trained only on observations (or benchmarks) of individual services, never on a run of the evaluated composition S. The ground truth is the measured $(1-\epsilon)$ quantile of the end-to-end latency across S.

- **No queueing (*GP Sum*, *DNN Sum*):** per service, a model predicts the execution time from the configuration; the end-to-end latency is the $(1-\epsilon)$ quantile of their sum. *GP Sum* reuses the *Pegasus*'s GPs, whereas *DNN Sum* uses a feed-forward network with the standard deviation of its training residuals. GP and neural-network regressors are common performance models of configurable systems [3], [11], [20], [21], [30]; summing their per-service predictions is the most direct way to compose their performance without data of the composition.
- **Queueing formula (*GP Queue*):** the same GP predictions enter the Allen–Cunneen approximation of the mean waiting time in a G/G/c queue [4], which reduces to Kingman's formula [18] for one worker, with an exponential waiting-time tail [16]; the per-service $(1-\epsilon)$ quantiles are added. *GP Queue* accounts for queueing, but it is an approximation, not a bound, and it uses GP predictions without a margin for epistemic uncertainty.
- **Learned quantiles (*Quantile Union*, *Conformal*):** the same GPs are used to predicts every service's $(1-\epsilon/K)$ latency quantile; in *Quantile Union*, the K predictions are added as a union bound. *Conformal* additionally multiplies each prediction by a split-conformal correction factor calibrated on held-out training configurations [32]. In contrast to *Pegasus*, both baselines need a load test per service and configuration, and assume that services see the same arrival process as during isolated training.

As a reference, an *Oracle* adds the *measured* $(1-\epsilon/K)$ latency quantiles of the services inside the composition. The *Oracle* uses the ground truth and is not a competitor; it shows the tightest result any union bound over services can reach.

Guarantee Tightness. Tightness is the predicted latency divided by the measured p95 of a run; below $1\times$, the SLO is violated. A run is covered if the prediction is not below the lower end of the 95% confidence interval of its measured p95. A method *holds* if, under every seed, it covers at least $\gamma = 95\%$ of the runs. We compare tightness only among methods that hold.

Computational Complexity. We measure the time to train the models of all services, and to evaluate all held-out runs of one seed. Pegasus, *GP Sum*, and *GP Queue* share the same GPs and thus the training time.

TODO: machine

B. Coverage and Tightness of Service Bounds

1) Base Case: We bound the p95 of every unthrottled run of the 15 chains (1,413 runs), with three seeds that each draw 25 benchmark runs per service. To test whether it matters where a service was observed, we repeat B1 with *leave-one-out* training: a service is trained on its runs in the other chains of its application. On *Vision* 1 (detector $\rightarrow$ transformer $\rightarrow$ compressor, 99 runs), which B2–C1 build on, we add *DNN Sum* and ablate six compositions of Pegasus on the same GPs.

a) Results: **Only Pegasus holds on all chains.** Fig. 3 shows that Pegasus Node and SNC hold on all 15 chains, at a median of $1.72\times$ and $2.72\times$. *GP Sum* fails on all chains, since it ignores queueing. *GP Queue* fails on 5 and *Quantile Union* on 4 chains; on *Vision* 1, *GP Queue* is tighter than Pegasus ($1.19\times$ vs. $1.58\times$). *Conformal* holds on all chains at $8.36\times$. With leave-one-out training, Pegasus still holds everywhere ($1.78\times$, $2.78\times$), while *Quantile Union* drops to 7 chains and *Conformal* to 13. On *Vision* 1, *DNN Sum* fails and all six compositions of Pegasus hold.

TODO: ablation numbers; base chain favors GP Queue

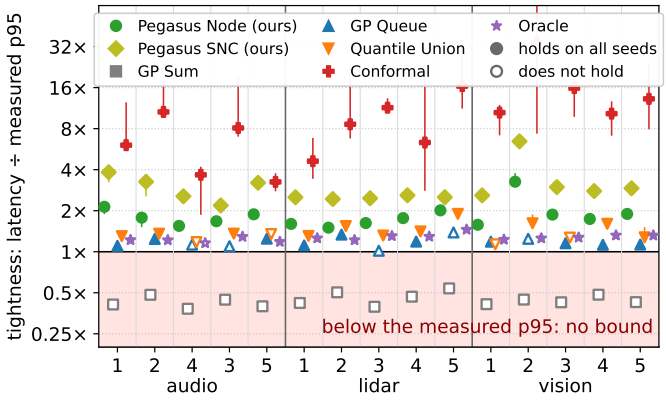


Fig. 3. B1: tightness per chain (median over seeds, bars: range); hollow markers do not hold.

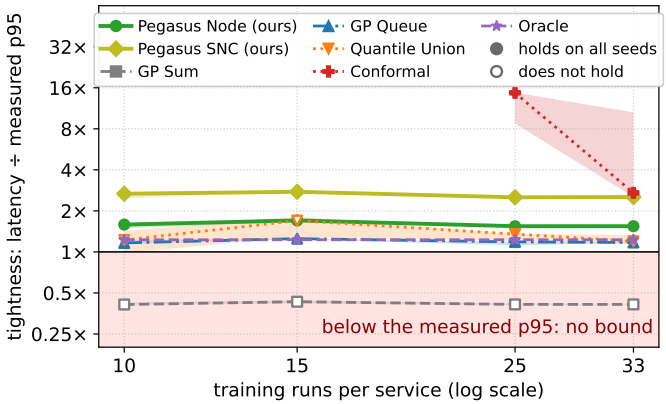


Fig. 4. B2: tightness on *Vision* 1 against benchmark runs per service.

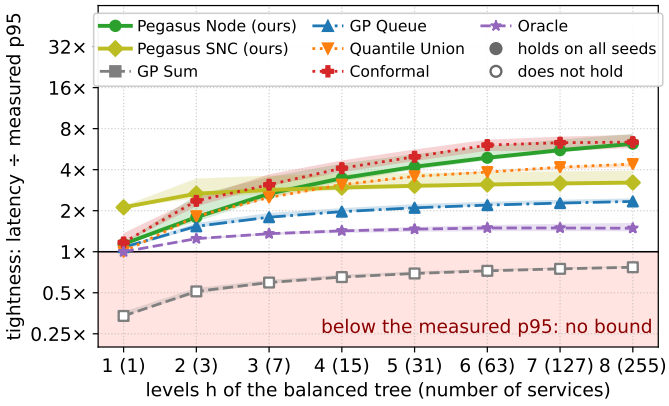
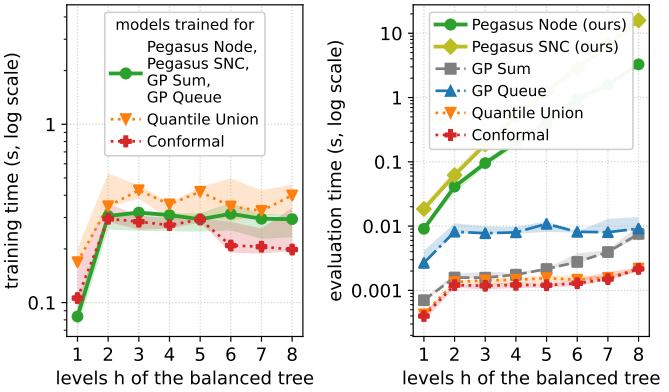


Fig. 5. B3: tightness against tree depth (synthetic).



(a) Training. (b) Evaluation.

Fig. 6. B3: computation time against tree depth (synthetic).

2) Varying Evidence per Service: On *Vision* 1, we train every service on 10, 15, 25, or 33 benchmark runs, drawn anew per size and seed; 33 is the most that every service has. *Conformal* needs at least 19 runs for calibration and is not computed below 25.

a) Results: **Ten benchmark runs per service suffice.** Fig. 4 shows that Pegasus holds at every size, with Pegasus Node at 1.59× to 1.54× and Pegasus SNC at 2.67× to 2.53×. *Quantile Union* fails at 10 and 15 runs; *Conformal* tightens from 14.7× to 2.7× between 25 and 33 runs. Training takes below 0.5 s for every method; evaluating 99 runs takes 28 ms (Node) and 55 ms (SNC). (*Range too narrow to show the epistemic margin.*)

TODO: start at 3–5 runs; B2b?

3) Varying Chain Depth: We simulate balanced fork-join trees with $h = 1, \ldots, 8$ levels ($2^h - 1$, up to 255 services). An item enters all leaves at once; each inner service waits for both children, then processes the item FIFO. Leaves alternate between two generated services, inner services use a third. Every tree runs at 50% utilization of its slowest service, with 100 held-out configurations per seed. Each method bounds every leaf-to-root path and takes the maximum. (*Synthetic.*)

TODO: real trees

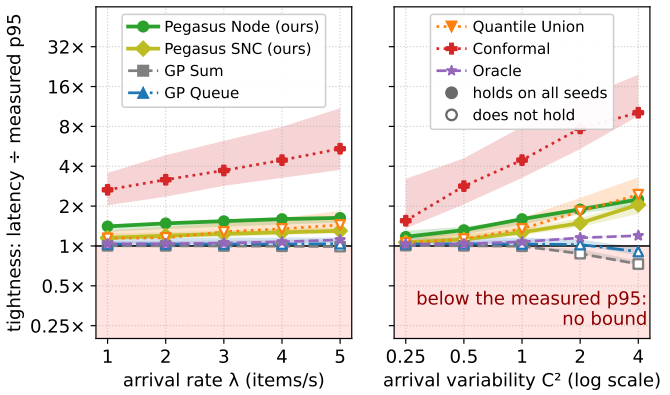
a) Results: **Pegasus SNC scales best with depth.** Fig. 5 shows that all methods but *GP Sum* hold up to 255 services. Pegasus Node loosens from 1.14× to 6.20×, since it splits ϵ across all services; Pegasus SNC pays ϵ once per path and only loosens from 2.12× to 3.22×. The measured p95 grows from 0.23 s to 0.78 s. Training takes 0.08 s to 0.29 s (Fig. 6). (*GP Queue and Quantile Union hold on all synthetic trees, unlike on real chains.*)

4) Varying Load and Arrival Pattern: On a simulated chain of three services, we vary (a) the Poisson arrival rate from 1 to 5 items/s and (b) at 4 items/s, the squared coefficient of variation of gamma-distributed gaps, $C_a^2 \in \{0.25, 0.5, 1, 2, 4\}$. The execution-time GPs stay unchanged; Pegasus receives the arrival process and *GP Queue* its rate and C_a^2. *Quantile Union* and *Conformal* are retrained for every setting. (*Synthetic; real runs have one fixed rate each.*)

a) Results: **Pegasus needs no retraining for a new workload.** In Fig. 7a, Pegasus holds at every rate; Pegasus Node loosens from 1.41× to 1.63× and Pegasus SNC from 1.15× to 1.30×, whereas the retrained *Conformal* grows from 2.66× to 5.43×. All other baselines fail. (*Median utilization of the busiest service only reaches 0.16.*)

TODO: utilization up to 0.9

Burstier arrivals widen the bound with the latency. In



(a) Arrival rate (items/s). (b) Arrival variability C_a^2.

Fig. 7. B4: tightness against load and arrival pattern (synthetic).

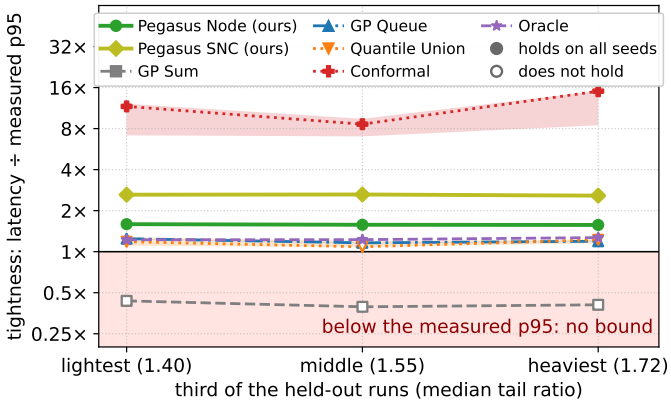


Fig. 8. B5: tightness on *Vision* 1 by tail ratio.

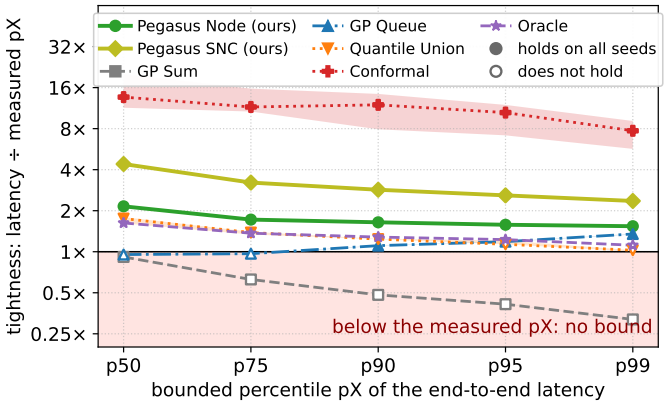


Fig. 9. B6: tightness on *Vision* 1 against the bounded percentile.

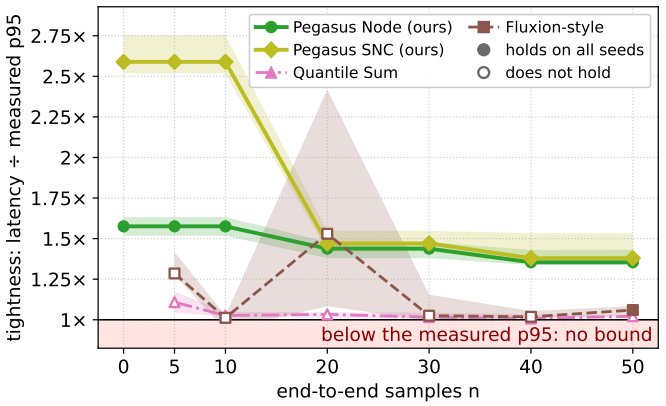


Fig. 10. C1: tightness on 49 test runs against the end-to-end samples n.

Fig. 7b, the measured p95 grows from 0.23 s to 0.34 s with C_a^2; Pegasus Node holds throughout and grows from 1.17× to 2.24×, *Conformal* from 1.55× to 10.15×. Pegasus SNC fails for $C_a^2 < 1$.

TODO: explain SNC for $C_a^2 < 1$

5) Varying Tail Behaviors: We sort the runs of *Vision* 1 by their tail ratio, the p99 divided by the median execution time at the service where this ratio is largest, and split them into thirds (median ratios 1.40, 1.55, 1.72). Models and predictions are those of B1. (*Tails barely differ, and the groups also differ in configuration.*)

TODO: heavier or synthetic tails

a) Results: **The tail does not affect Pegasus.** Fig. 8 shows that Pegasus, *GP Queue*, and *Conformal* hold in all groups. Pegasus Node stays at 1.59× to 1.57×, while *Conformal* grows from 11.6× to 15.1×. *Quantile Union* fails for the two heavier groups.

6) Varying Percentiles: On *Vision* 1, we bound the p50, p75, p90, p95, and p99, i.e., $\epsilon = 1-$ level. Pegasus and *GP Queue* reuse the models of B1; *Quantile Union* and *Conformal* train one model per level.

a) Results: **Pegasus is tightest for strict SLOs.** Fig. 9 shows that only Pegasus and *Conformal* hold at all levels. Pegasus Node tightens from 2.15× at p50 to 1.54× at p99, Pegasus SNC from 4.40× to 2.36×. *GP Queue* fails at p50 and p75, *Quantile Union* at p95 and p99. (*A run's p99 rests on few items.*)

C. Iterative Optimization of Service Bounds

1) Fine-Tuning Bounds with E2E Samples: We split the 99 runs of *Vision* 1 into up to $n = 50$ end-to-end samples and 49 test runs. Pegasus multiplies its bound by the $\lceil (n+1)\gamma \rceil$-th smallest ratio of measured to predicted p95 among the samples (split-conformal), which needs $n \geq 19$. As baselines trained on the samples, *Quantile Sum* adds the per-service p95 measured inside the chain, and a *Fluxion*-style model [21] chains one GP per service that takes the chain configuration and the predicted quantiles of the preceding service.

TODO: conformal step in Sec. IV; no retraining vs. Fluxion

a) Results: **End-to-end samples tighten Pegasus without breaking it.** Fig. 10 shows that Pegasus holds for every n. From $n = 20$, the calibration takes effect: at $n = 50$, Pegasus Node tightens from 1.58× to 1.35× and Pegasus SNC from 2.59× to 1.38×. *Quantile Sum* never holds, the *Fluxion*-style model only at $n = 50$.

2) Inferring Faster Service Compositions: As in Fluxion [21], each method guides an optimizer through 100 candi-

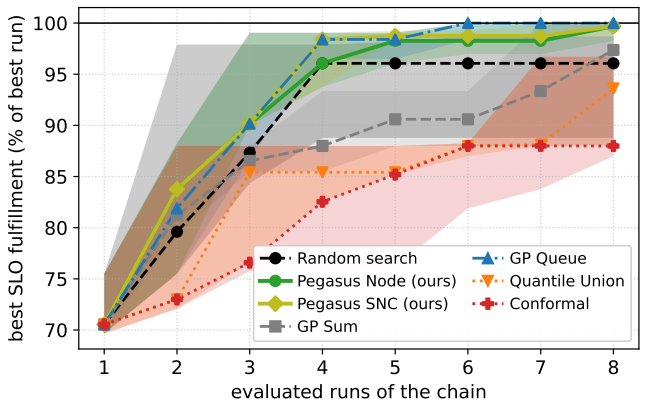


Fig. 11. C2: best SLO fulfillment found against evaluated runs (gradient steps, synthetic).

date configurations of a simulated chain (Poisson, 4 items/s). After each of eight evaluated runs, the method refits its models. The objective is the weighted SLO fulfillment of [29] over a 251 ms latency SLO and quality SLOs. *Best of the runs* picks the candidate with the highest predicted fulfillment; *gradient steps* moves along its predicted gradient to the nearest candidate. Random search is the reference. (*Synthetic; real runs lack a common rate.*)

TODO: min. budget objective

a) Results: **Pegasus keeps its promises.** Fig. 11 shows that, with gradient steps, Pegasus and *GP Queue* reach the best fulfillment within eight runs, ahead of *GP Sum* (97%), *Quantile Union* (94%), and *Conformal* (88%). Every configuration that Pegasus predicted to meet the latency SLO did; *GP Sum* broke 36% of such predictions. (*With best of the runs, nearly all methods find the best candidate.*)

VI. Discussion

@Víctor most likely

A. Limitations

Heavy-Tailed Execution Times. The performance of distributed systems is known to show heavy-tailed behavior, e.g., during garbage collection. *Pegasus* requires sub-Gaussian execution times (Assumption 4). If a Gaussian is fitted to a service with a heavier tail, the bound stays accurate where queueing dominates the latency, because a long queue depends mainly on the mean and the variance of the execution time; it becomes optimistic where the tail of a single execution dominates, i.e., at low utilization, for very regular arrivals, and for high percentiles. The per-service composition is affected more strongly in deep pipelines, as it bounds every service at ϵ/K.

Two extensions keep the framework applicable. First, services in production enforce timeouts; an execution time that is cut off at a timeout τ is bounded and therefore sub-Gaussian, so the guarantee holds for the service as deployed.

TODO: Draft of the mechanism: with a hard timeout, $Y_i \leq \tau$ always, and Bennett's inequality bounds the MGF from the variance and the range, which is tighter than a Gaussian with the range as scale. Without a hard timeout: cap at a high quantile and reserve a part of ϵ for items beyond the cap; this needs a proof for the items in the same busy period.

Second, network calculus has been extended to heavy-tailed processes with power-law instead of exponential envelopes [22]; the resulting bounds decay polynomially, are considerably looser, and lose the closed forms that make *Pegasus* fast to evaluate, which is why we do not adopt them.

Sampling from Evaluation Rounds. The idea was that we can drastically simplify experiments by not propagating decisions (e.g., *where* to explore) back to the services, but instead use the grading function to sample from all conducted runs. Given the coverage of the solution space, we can bound the inaccuracy this creates by finding the largest distance between two points, or a point and the parameter bounds.

VII. Conclusion & Future Work

@Víctor most likely

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POTENTIAL APPENDIX

1) Description of the service generator.
2) List of the services and their states and configuration knobs in more detail. This can be clusters of service types, as for *Fluxion* in [21].
3) Description of the code artifact, or at least a pointer to the extended description in the README there.